

PrimeFocus

January 2026



WHEN

January 16, 2026

Doors open at 7:00pm

Meeting starts at 7:30pm

Lecture starts at 8:00pm

WHERE

Unitarian Church

1893 North Vasco Road

Livermore, CA 945571

and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

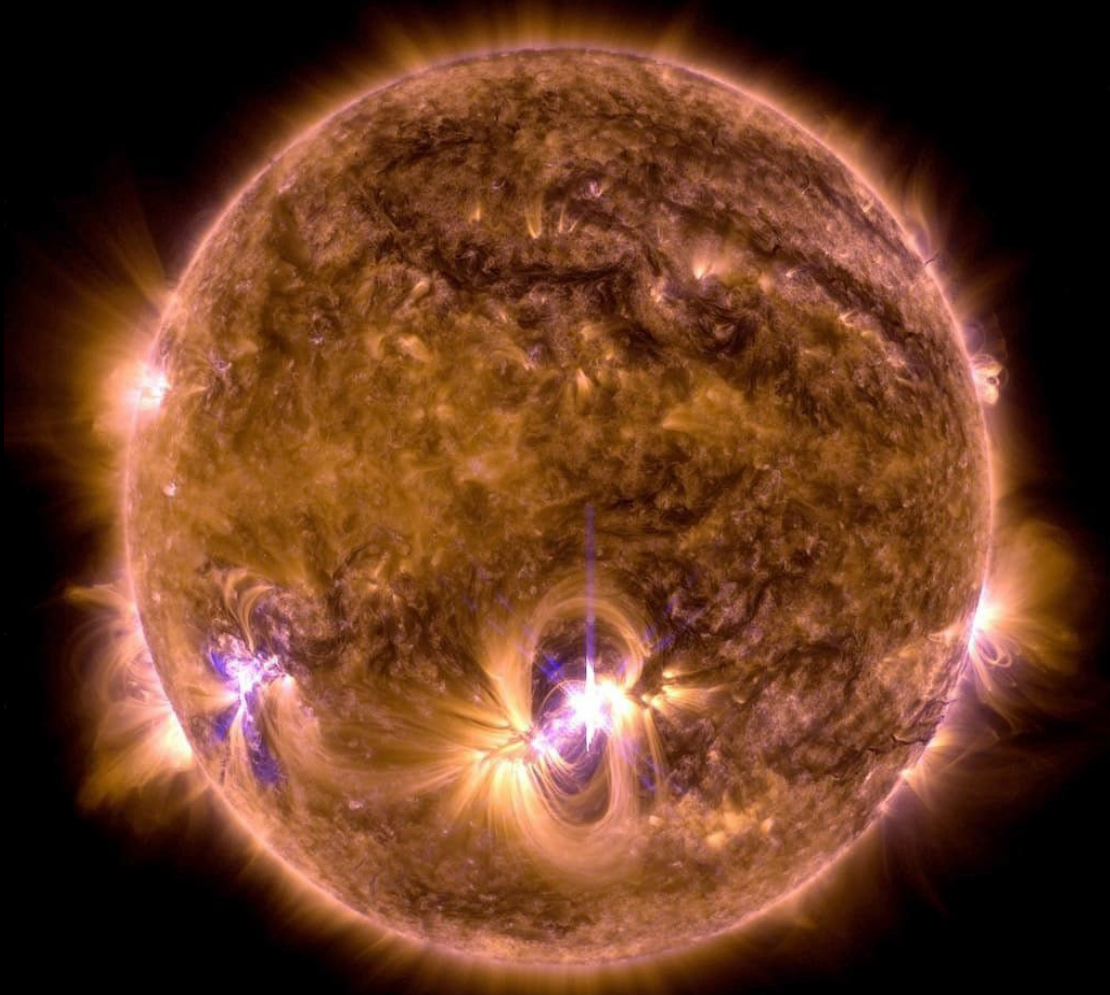
Stars Without Nuclear Fusion - Much of the Physics Without all of the Confusion **Dr. Kirk T. Korista**

The phenomenon of nuclear fusion influences the structure and evolution of stars in many important ways, but several key physical processes in and behaviors of stars are ubiquitously mis-attributed to nuclear fusion. These serve as major misconceptions that lead to widespread confusion about how stars work:

1. Nuclear fusion does not support stars against the force of gravity and gravitational collapse, nor does it generate significant radiation pressure to do so as is mysteriously invoked. In fact, radiation pressure rarely contributes significantly to supporting stars at all.
2. Nuclear fusion is not the reason stars are hot.
3. Nuclear fusion is not the reason stars are luminous, nor does it control directly the star's luminosity.

None of this is new physics. We do not have a problem with our models; we have a major problem in how we translate them into the English language and communicate them to humans. Worse, some of the misconceptions are spreading into the peer-reviewed literature.

In point of fact, stars do not require nuclear fusion in order to exist as hot, luminous objects that are in force balance against gravity. In order to illuminate their fundamental natures, we explore stars without nuclear processes and compare their evolutionary behavior to stars with nuclear fusion.



Our Sun, a main sequence star, emits a strong solar flare flashes in this image captured by NASA's Solar Dynamics Observatory.
NASA/SDO

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
January 16	January 19	January 3
February 20	February 23	February 7
March 20	March 23	March 7

Money Matters

As of the last Treasurer's Report on 12/22/25, our club's account balance is \$50,696.43, this includes \$12,705.46 in the H2O Rebuild fund.

TVS Welcomes New Members

TVS welcomes new member Violet Bickham. Please say hello and chat with them during our meetings.

2026 TVS Club Star Party Schedule

Save the dates for the 2026 Club Star Parties.

Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

No events currently scheduled for Del Valle.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

No events currently scheduled for Tesla Vintners.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

No events currently scheduled for H2O.

January 14: School Star Party Lawrence Elementary School Livermore, 2451 Portola Ave, Livermore. Set-up 5:50pm, Observing from 6:30-7:30pm.

January 20: School star party at St. Michael's School, 345 Church Street, Livermore. Set-up 5:30pm, Observing from 6:30 - 8:00pm

January 22: School star party at Dublin Elementary School, 7997 Vomic Road, Dublin. Set-up 5:00pm, Observing from 6:00 - 8:00pm

January 27: Family Science Night Marilyn STEAM School, 800 Marylin Ave, Livermore. Set-up 5:30pm, Observing from 6:00 - 8:00pm

CALENDAR OF EVENTS



January 16, 17, 23,24, 30, 31 February 6, 7,13, 14, 7:30-10:30pm

What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot's historic telescopes. Chabot's three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center's 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see
<https://chabotspace.org/events/events-listing/>
<https://eastbayastro.org/chabot-telescope-status/>

January 28, 7:00pm

What	Pictures of Distant Worlds
Who	Silicon Valley Astronomy Lecture Series
Where	Smithwick Theater at Foothill College, in Los Altos
Cost	Free

In the past three decades, more than 6000 planets have been

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discovered orbiting other stars beyond our own Solar System. However, we still don't know if our Solar System is rare or unique — the powerful techniques that detect extrasolar planets have discovered systems very different than our own. In recent years, advances in technology have allowed a handful of giant planets around other stars to be imaged directly. Find out about the first-ever images of other solar systems — and the technology that has allowed us to discover them, such as the Gemini Planet Imager — as well as the future planet-hunting space telescopes. The ultimate goal is detection of a second 'pale blue dot' — an Earth twin where we could even see the biosignatures of extrasolar life.

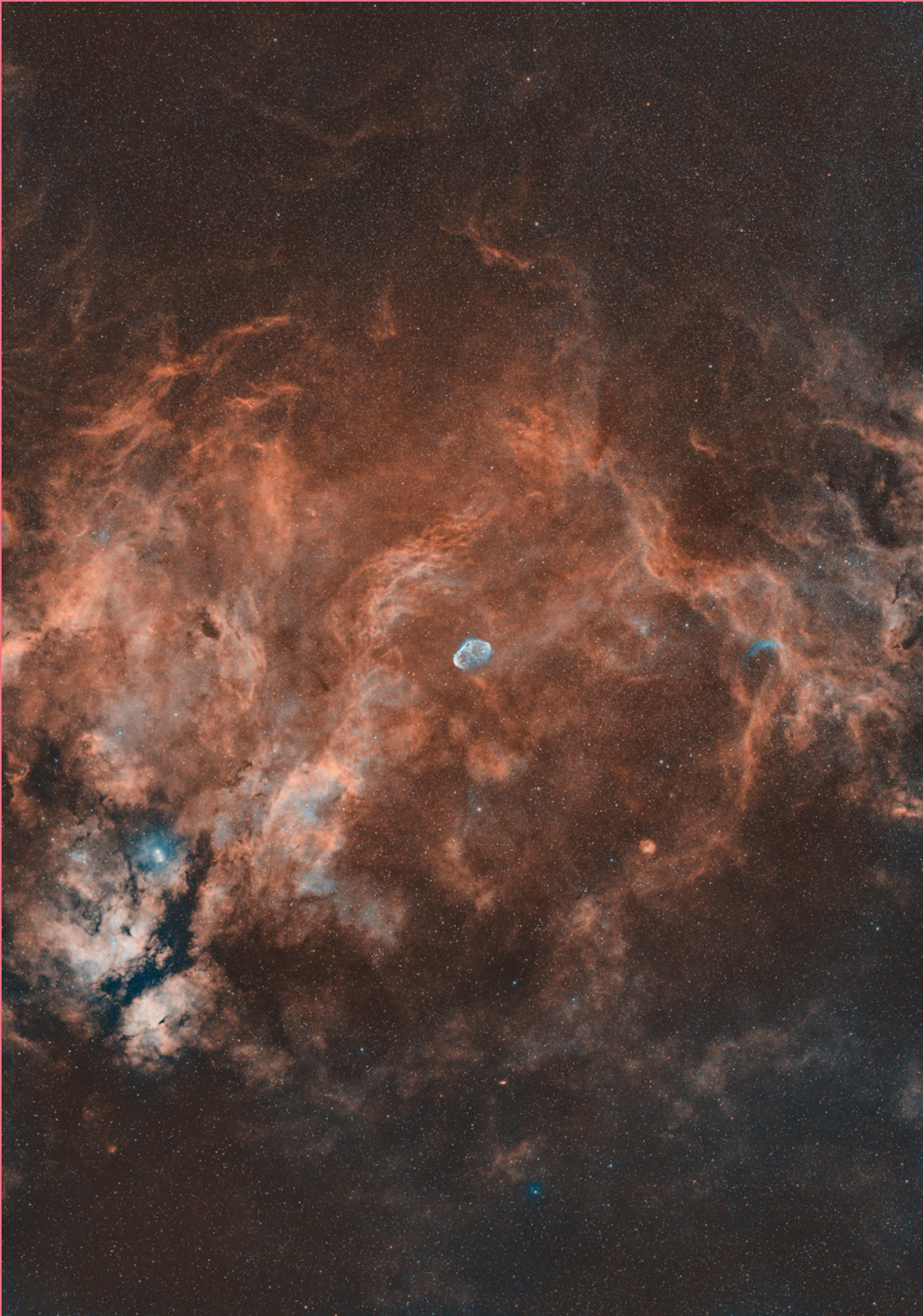


Gemini Planet Imager (GPI) on Gemini South telescope. Credit: International Gemini Observatory/NOIRLab/NSF/AURA/M. Paredes

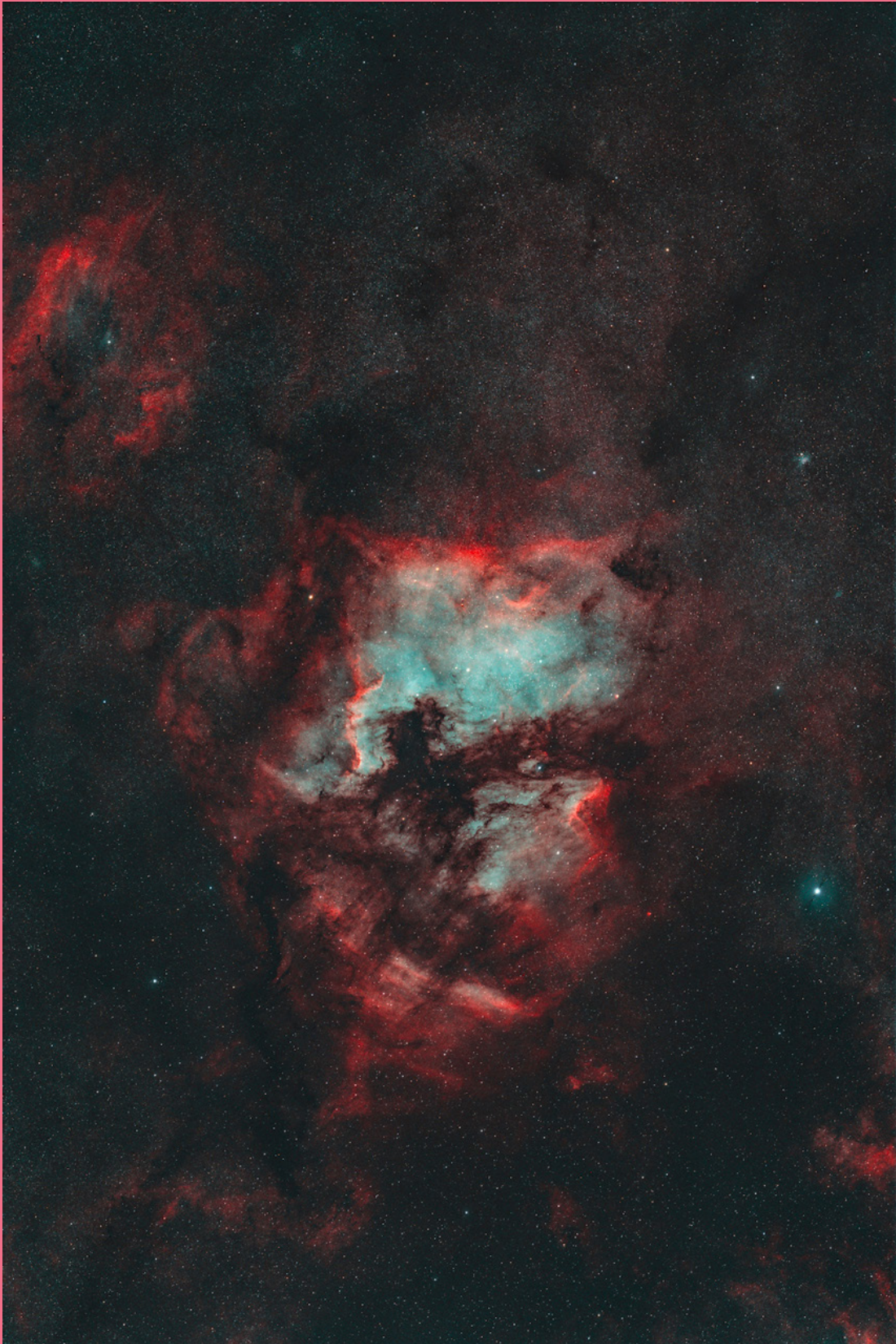
Bruce Macintosh is the Director of the University of California Observatories in California and Hawaii. His research focuses on the study of extrasolar planets, in particular the study of such planets through direct imaging. Direct imaging of extrasolar planets involves blocking, suppressing, and subtracting the light of the bright parent star so that a planet hundreds of thousands of times fainter can be seen and studied in detail. Dr. Macintosh co-led the team that imaged the first extrasolar planets, and was the Principal Investigator of the Gemini Planet Imager, an advanced adaptive optics planet-finder for the Gemini South telescope.



For parking information, see <https://foothill.edu/parking/>



NGC 6888, By Kevin McLoughlin.
For full size image,see <https://app.astrobin.com/i/bv3vrg>



North America and Pelican Nebulas, By Kevin McLoughlin.
For full size image,see <https://app.astrobin.com/i/4akrnp>



The Moon, By Jayson Martin.

WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

January 2026

18	Sun	New Moon
22	Thu	At dusk, looking south-southwest, the Moon is just 6° lower right of Saturn
26	Mon	Moon is at first quarter; Algol shines at minimum brightness between 9:58 and 11:58 PM
27	Tue	Moon sits left of the Pleiades
30	Fri	At dusk, Moon, Jupiter and Pollux form a triangle while looking east. Jupiter is just 3½° lower right of the moon

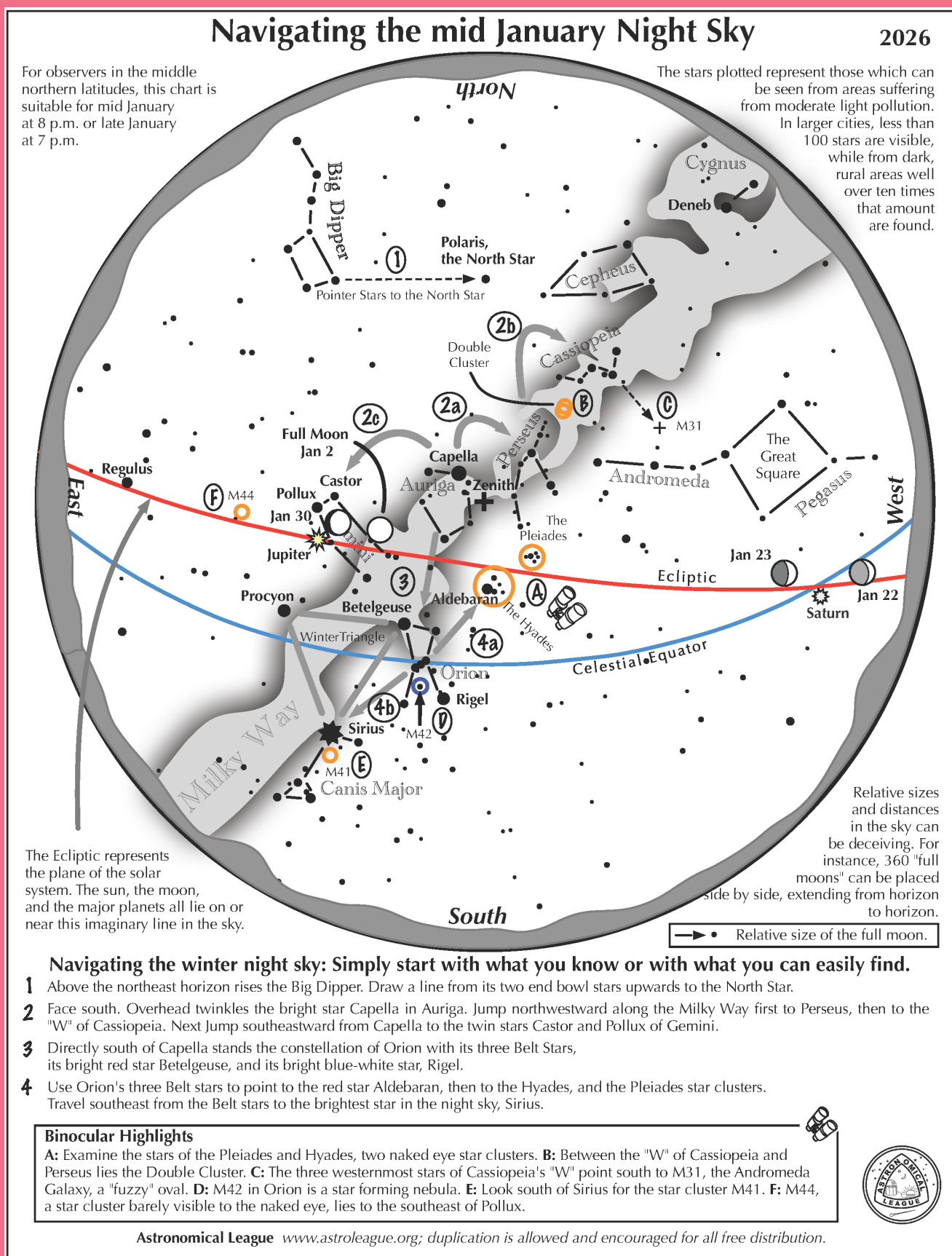
February 2026

1	Sun	Full Moon
2	Mon	Watch the moon rise in the east with Regulus, As the pair rise the moon occults the star.
6	Fri	Looking east, the moon and Spica are just 1½° apart
9	Mon	Moon is at last quarter
11	Wed	In the morning, low in the southeast, Moon is just 3¼° upper right of Antares
17	Tue	New Moon

OFFICERS AND VOLUNTEER POSITIONS

Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
Vice-President Aris Pope vice_president@trivalleystargazers.org	Club Star Party Coordinator Johnathan Bailey coordinator@trivalleystargazers.org	Webmaster Swaroop Shere webmaster@trivalleystargazers.org	Program Coordinator Ron Kane programs@trivalleystargazers.org
Treasurer Ernest Epley treasurer@trivalleystargazers.org	Del Valle Coordinator David Wright delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H2O@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Johnathan Bailey outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR JANUARY 2026





Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.